



NFC BUSINESS CARD With Energy Harvesting

ASHWINI KUMAR SINHA

NFC (near field communication) is a short-range wireless technology that enables data exchange between two devices placed within 4cm of each other. In an NFC business card, a built-in NFC chip lets users instantly access contact details or a website by simply tapping the card with an NFC-enabled smartphone.

This system presents a universal NFC smart card based on the ST25DV64K-IER6T3 dynamic NFC/RFID tag IC. The versatile IC enables the card to function as a professional business card, employee ID badge, NFC memory storage device, or secure access token. However, it is not intended for certified banking or ATM applications because of regulatory and certification requirements.

The ST25DV64K-IER6T3 features 64-kbit (8kB) user EEPROM memory, which can be divided

into up to four flexible, password-protected areas. It supports dual interfaces—ISO 15693/NFC Forum Type 5 for smartphone compatibility and a high-speed I²C interface operating at up to 1MHz. The IC also incorporates fast transfer mode (FTM) with a 256-byte mailbox for efficient data exchange between the RF and I²C interfaces, along with extensive configuration options for a wide range of NFC applications.

Although the primary objective of this system is to create an NFC business card for convenient sharing of vCard and NDEF messages, the card also supports energy harvesting through the V_{EH} pin. This feature can deliver approximately 2mW-3mW of power under typical conditions and up to 10mW-20mW under optimal conditions, making it suitable for powering low-current devices such as LED indicators without an external power source.

Features such as a programmable GPO interrupt, multiple 64-bit password-protected memory areas, 1 million write cycles, and 40-year data retention make the ST25DV64K-IER6T3 a reliable and future-proof platform for developing smart, interactive NFC cards. The prototype

of the NFC smart card is shown in Fig. 1. The components required to build the system are listed in the Bill of Materials (Table 1).

Circuit design

For an NFC business card, two key aspects require careful consideration: energy harvesting and antenna design.

Antenna design is critical in any NFC-based application. Best practices, design guidelines, and design tools are beyond the scope of this article and will be covered separately. Additional guidance can also be found in online technical resources. The discussion here focuses on selecting an appropriate NFC IC and the corresponding circuit design.

The ST25DV64K-IER6T3 dynamic NFC/RFID tag IC was selected for this system. Although several alternatives are available from NXP, including the NTAG and ICODE series, any NFC IC that provides a suitable balance of memory capacity, dual-interface communication (RF and I²C), and energy harvesting capability can be used. The ST25DV64K-IER6T3 offers an excellent combination of these features, making it well

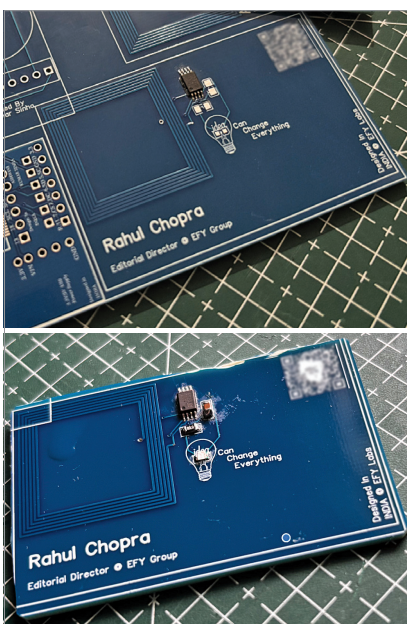


Fig. 1: Author's prototype of NFC business card

TABLE 1
BILL OF MATERIALS

Component	Designator	Footprint	Quantity	Manufacturer Part Number	Manufacturer
10 μ F capacitor	C1	0805	1	CL10A106KP8NNNC	Samsung
2.7k Ω resistor	R1	0805	1	0805W8F2701T5E	Uni-Royal
NFC dynamic tag IC	U1	TSSOP-8 (4.4mm $\times$ 3.0mm, 0.65mm pitch)	1	ST25DV64K-IER6T3	STMicroelectronics
Red LED	U2	LED0603	1	KP-1608SURCK	Kingbright
NFC antenna	U3	PCB spiral antenna	1	Integrated on PCB	—